

Fault-tolerant Preparation of a Logical Qubit in Diamond Processors

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We demonstrate a simulated protocol for fault-tolerant preparation of the logical state $|-\rangle_L$ in a diamond NV-center processor. Building on the five-qubit $[[5, 1, 3]]$ stabilizer code, we employ an NV electron spin as an ancilla and five surrounding ^{13}C nuclear spins as data qubits, supplemented by a ^{14}N flag qubit and “record” qubits to replicate real-time feedforward. Using `Qiskit` simulations, we compare non-fault-tolerant and fault-tolerant circuits under randomized Pauli errors on the ancilla. Over 1000 trials, the fault-tolerant scheme achieves a logical-state preparation success rate of 79.0%, more than double the 39.6% observed for the non-fault-tolerant approach. These results illustrate the practical benefit of fault-tolerant design in solid-state quantum registers.

I. INTRODUCTION

The manuscript is structured in the following order. In section II, we write the contribution from each member of the group. We cover important background knowledge in section III, introduce our circuit and methods to prepare $|-\rangle_L$ in section IV, verify the fidelity of $|-\rangle_L$ in section V, and draw overall conclusions in section 6 in section VI

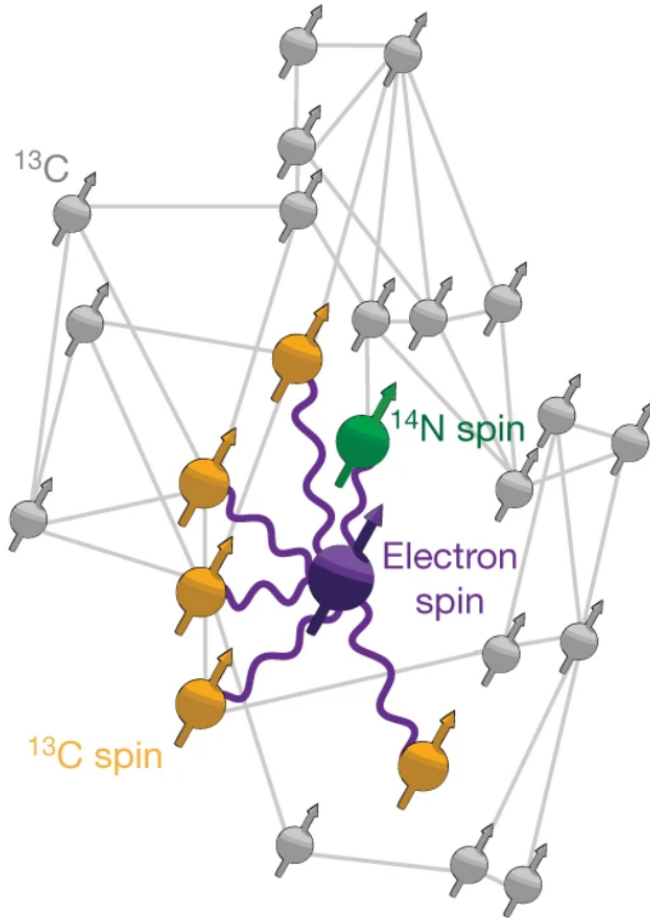


FIG. 1. An illustration of the 7-qubit NV system in the diamond lattice. Reproduced from [1].

II. CONTRIBUTION STATEMENTS

- Alex - Carried out the simulation that tested the validity of the Fault-Tolerant preparation of the logical state. Worked on top of the circuits prepared by Dex and used the help of record qubits to achieve an equivalency of real-time controller that help capture error in the Fault Tolerant preparation. Examined the simulated state vector after `qiskit` simulations in the form of density matrices and analyzed desired parts such as the data bits and the record bits with reduced density matrices. Measured the pauli operators of the prepared state for both the non-FT and FT preparation to test the integrity of the prepared state against the ideal state. Ran simulations for 1000 iteration to get success rate of preparing desired logical state to obtain the success rate for the two preparations. All of Alex's work on coding can be found in [Paper Figures-alex.ipynb](#) on the GitHub link attached in the Appendix (VII).
- Dex - Helped make the circuit for the non-fault-tolerant and fault-tolerant stage of our error-correcting protocols. Initially proposed finding statevectors or using probabilistic counts from `Qiskit` [4] to explain the functioning of the circuit. This idea was switched with Alex's approach for better visualization. All of Dex's work is visible in the 'Paper Figures' notebook found on the GitHub link attached in the Appendix (VII). Also, helped with the design of the poster and formatting of this report.
- Saahit - Identified the topic of interest. Focused on the application of the explored system in NV center defects, i.e., focused on the theoretical component, including the readout procedures, experimental setup, and the physical principles of NV center qubits. Assisted Dex in the construction of the `Qiskit` circuit for the non-fault-tolerant stage of the error-correcting protocols. Contributed to the formatting and content of the report and poster.

III. BACKGROUND

A. Nitrogen Vacancy Centers

A nitrogen-vacancy (NV) center in diamond consists of a nitrogen atom (replacing the usual carbon) adjacent to a vacancy in the diamond lattice. When the defect is in its negatively charged state (NV^-), it hosts an electronic spin triplet ground state, with spin projections $m = 0, \pm 1$. Even in the absence of an external magnetic field, there is a zero-field splitting of approximately 2.87 GHz between the $m = 0$ and $m = \pm 1$ states, due to spin-spin interactions within the defect.

A key feature of NV centers is that their spin state influences their optical behavior. When excited with green (532 nm) light, the NV center fluoresces red (637 nm), but the intensity of this fluorescence depends on the spin state. Specifically, the $m = 0$ state tends to emit more fluorescence than the $m = \pm 1$ states, which are more likely to undergo non-radiative decay through an intermediate singlet state. This difference enables optical readout of the spin state, forming the basis for many quantum applications of NV centers [2].

Coherent control is achieved using microwave pulses to drive transitions between the $m_s = 0$ and $m_s = 1$ states, with Rabi frequencies typically around 15 MHz. In experiments like [1], this allows high-fidelity single-qubit gates. Nuclear spins near the NV—such as the ^{14}N intrinsic spin and nearby ^{13}C isotopes—are coupled via hyperfine interaction and controlled with RF pulses. Together with optical readout and initialization, these tools provide a full set of quantum control operations [3]. At cryogenic temperatures, resonant optical excitation can further enable single-shot readout of the NV spin, though many NV-based experiments still operate at or near room temperature.

NV centers are distinguished by their exceptionally long coherence times. The electron spin T_2 can exceed 1 ms in ultrapure diamond and be extended to seconds with dynamical decoupling. Nuclear spins (e.g., ^{13}C) offer even longer coherence, acting as quantum memories. For instance, [1] reports T_2 values around 1.18 ms at 4 K, with T_1 lifetimes exceeding 10^3 s. These long timescales, combined with fast gate operations, allow many error-corrected operations within the coherence window [1].

The strong, controllable coupling between the NV electron and nearby nuclear spins enables multi-qubit registers within a single defect [1].

As seen in Figure 1, five ^{13}C spins were used as data qubits, with the ^{14}N spin as a flag qubit and the NV electron as an ancilla for syndrome extraction. The electron can be repeatedly entangled, measured optically, and reused without disturbing the nuclear qubits, enabling fast, non-destructive stabilizer measurements.

This architecture allowed for the realization of a five-qubit quantum error-correcting code with feedforward-based logical operations and stabilizer readout.

B. Error Correcting Codes

Quantum computation works by controlling quantum analogs of classical bits called qubits. However, a physical system to support any qubit technology is susceptible to external noise and errors. In classical computers, classical bits are realized as an “on” or “off” state of the transistor made up of billions of electrons, making it easier to detect and eradicate high-error classical bits. In quantum computers, qubits are fragile quantum systems without protection from external noise, demanding the need for error correction protocols.

Stabilizer codes are a type of code used for quantum error correction. In this experiment, we use the $[[5, 1, 3]]$ stabilizer code, which means one logical qubit is encoded into five physical qubits. It is the shortest distance-3 ($d = 3$) code, which means it can correct unknown errors on $\frac{(d-1)}{2}$ physical qubits. It is defined by four independent stabilizers, which are the syndrome measurements.

Any logical state in the $[[5, 1, 3]]$ code is the $+1$ eigenstate of the four stabilizers $XXYIY$, $YXXYI$, $IYXXY$, $YIYXX$ with logical operators $XXXXX$ and $ZZZZZ$. The unique 4-bit syndromes between a single data qubit error and the stabilizers can be used to identify and correct errors.

In our circuit, we also add an auxiliary and flag qubit to support a fault-tolerant error correction code.

C. Pauli Errors Introduced

In this experiment, we will eventually investigate the impact of Pauli Errors on our $[[5, 1, 3]]$ code. The Pauli Errors we look at are the X , Y and Z errors, represented by 2×2 Pauli matrices described in Equation 1.

$$\begin{aligned} X &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ Y &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ Z &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned} \quad (1)$$

Therefore, suppose we have two bits $|0\rangle$ and $|1\rangle$. Then, $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$. Errors introduced by the X Pauli operators are bit-flip errors.

Similarly, $Z|0\rangle = |1\rangle$ and $Z|1\rangle = -|0\rangle$. Errors introduced by the Z Pauli operators are phase-flip errors.

Lastly, $Y|0\rangle = i|1\rangle$ and $Y|1\rangle = -i|0\rangle$. The Y Pauli operator acts as a bit-flip and a phase-flip error.

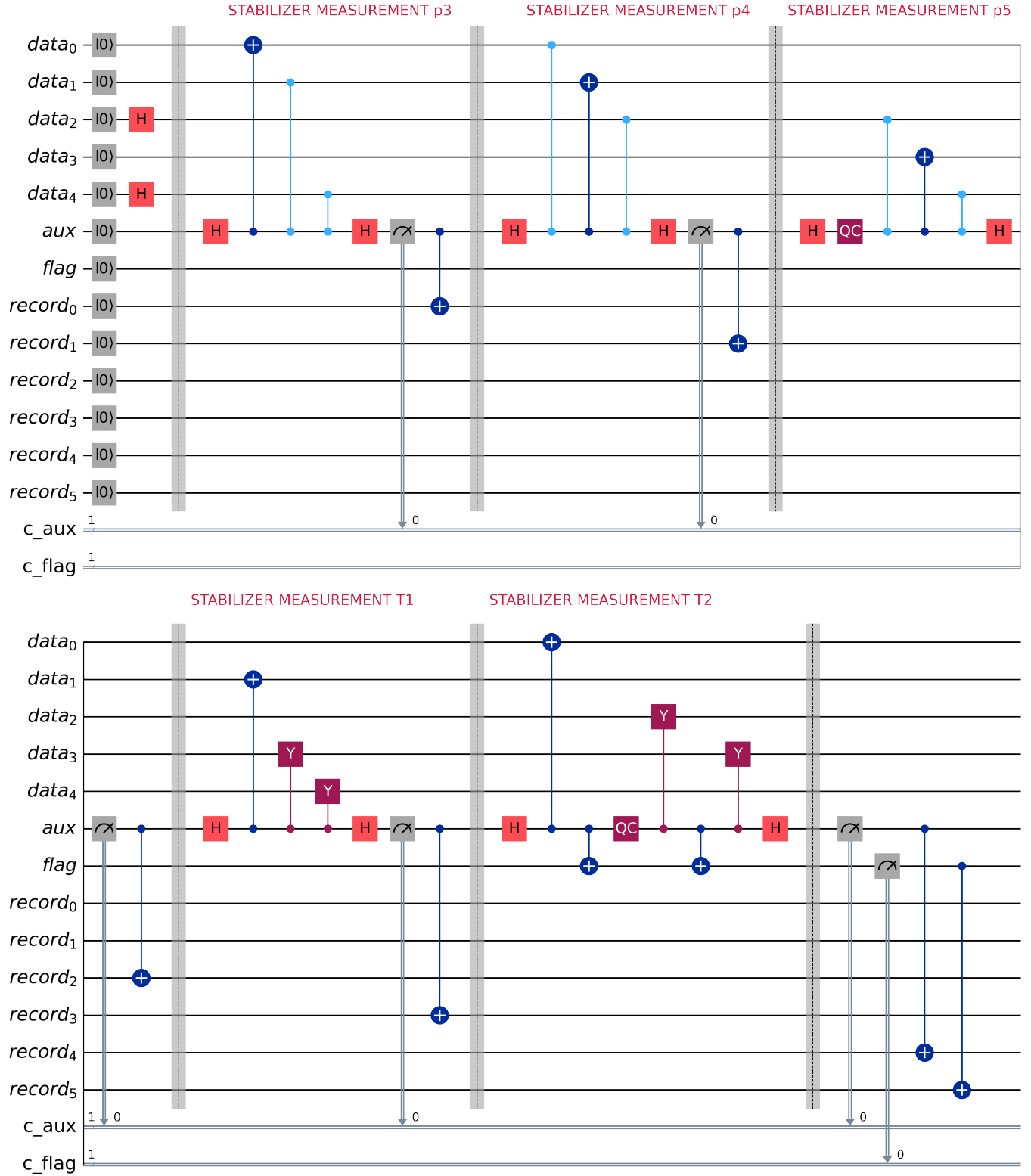


FIG. 2. (Top) The non-FT stage with the record qubits serving as a substitute for a real-time controller. Each of the classical measurements of the auxiliary bit is coupled with a record bit through a CNOT gate such that the record bits store the result of each of the classical measurements. (Bottom) The FT stage with the record qubits. Two quantum gates representing a random one-bit Pauli error are applied on an auxiliary bit in the non-FT and FT stages, respectively.

IV. METHODS

A. Logical Preparation of $|-\rangle_L$

In this experiment, we explore a five-qubit quantum error-correcting code. We create this circuit using *Qiskit* - a publicly available computational package on Python that simulates quantum circuits.

These five primary qubits carrying information are labeled “data qubits” in Figure 2. In addition to the data qubits, we use an auxiliary qubit to perform repeated stabilizer measurements on the data qubits and check for errors. We also use a flag qubit to detect possible errors on the auxiliary qubit. Our measurements on the auxiliary and flag qubits are recorded using two classical bits (labelled “c_aux” and “c_flag”).

Furthermore, to allow for non-destructive stabilizer measurements, we use “record qubits”. Since this experiment was purely based on simulations with *Qiskit*, we used our record qubits as a substitute for a real-time controller used to correct errors. We explore the purpose of record qubits further in section V.

In total, we use 13 qubits (five data qubits, six record qubits, one auxiliary qubit, and one flag qubit).

In the circuit demonstrated in Figure 2, we demonstrate the fault-tolerant preparation of the logical state $|-\rangle_L = \frac{1}{\sqrt{2}}(|0\rangle_L - |1\rangle_L)$. We begin by initializing our five data qubits as $|00+0+\rangle$.

The initial state $|00+0+\rangle$ passes through a series of stabilizer measurements that check for errors. Any single-qubit error on the data qubits during this stage corresponds to a unique five-bit syndrome. The measurement of the auxiliary qubit captures the eigenvalues of the five stabilizers used. These measurements control the successful preparation of the logical state.

Moreover, our circuit is divided into a non-fault-tolerant (non-FT) and a fault-tolerant (FT) stage. This division of our circuits into two primary stages is conveyed in Figure 2. In the non-FT stage, an error on the auxiliary qubit can cause weight-two errors, introducing logical errors to the state $|-\rangle_L$. However, in the FT stage, the stabilizer operators are arranged so that fault tolerance is ensured.

$|-\rangle_L$ is the +1 eigenstate of $p_1 = IZXZI$, $p_2 = ZIIZX$, $p_3 = XZIIZ$, $p_4 = ZXZII$, $p_5 = IIZXZ$, hence the stabilizer measurements we make in the non-FT and FT stages are permutations from p_1, p_2, p_3, p_4, p_5 . We also note that our initial state $|00+0+\rangle$ is the +1 eigenstate of p_1 and p_2 . Therefore, we only make stabilizer measurements p_3, p_4, p_5 in the non-FT stage. In the FT stage, we make stabilizer measurements $T_1 = p_2p_4p_5$ and $T_2 = p_1p_3p_5$.

Let the measurement associated with stabilizer p_i be m_i and let the measurement associated with stabilizer T_j be m_{T_j} where $i \in \{1, 2, 3\}$ and $j \in \{1, 2\}$. We claim that the state $|-\rangle_L$ is successfully prepared if:

- $m_{T_1} = m_2m_4m_5$

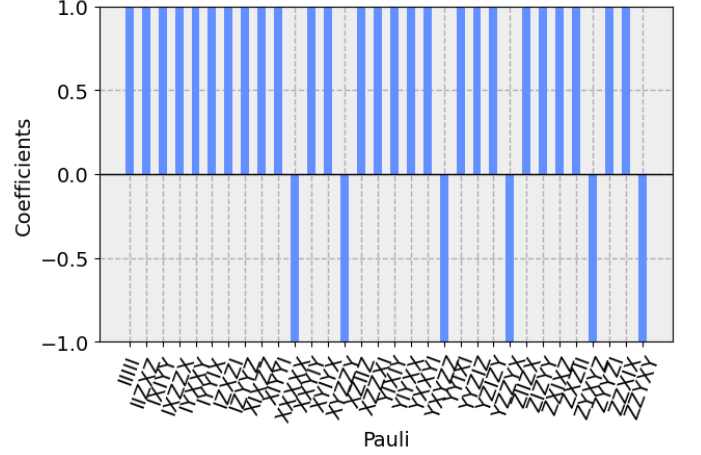


FIG. 3. The expectation of the Pauli operators of the prepared state from the non-FT stage without any error and with the real-time control accomplished with the record bits. We see that the prepares state is indeed the +1 eigenstate of $p_1 = IZXZI$, $p_2 = ZIIZX$, $p_3 = XZIIZ$, $p_4 = ZXZII$, $p_5 = IIZXZ$

- $m_{T_2} = m_1m_3m_5$
- The flag is $|0\rangle$, we prepare $|-\rangle_L$ successfully.

V. ANALYSIS

A. Qiskit Simulation of the Fault-Tolerant Circuit

Following the discussion on the validity of the fault-tolerant preparation of $|-\rangle_L$, we verified the integrity of the fault-tolerant preparation using *Qiskit*.

B. Preparing with Real-time Control and Verifying $|-\rangle_L$

Before diving into the fault-tolerance preparation of the circuit, we first verify that in the non-FT stage, the circuit indeed prepares the desired logical state when there's no error present.

The real-time controller was an essential part of successfully preparing the desired logical state in the experimental setup. At each stage of the classical measurement on the auxiliary bit, if the measured auxiliary bit is $|1\rangle$, meaning that the data bits are not in the +1 eigenstate of each measured stabilizer, the real-time controller restarts the circuit.

The preparation stage finishes when all the classical measurements of the auxiliary bit yield $|0\rangle$, which guarantees that the resulting state is the simultaneous +1 eigenstate of all measured stabilizers.

In *Qiskit*, however, the simulator that finds the resulting state of a circuit only records the state of the quantum

qubits. The result of each stabilizer measurement needs to be stored in classical bits. These measurements are rendered inaccessible after the simulation. Additionally, the simulator in `Qiskit` that allows for real-time feed-forward control must be connected to an IBM server, which is not supported by any fake backend we can replicate purely via simulations. Therefore, we introduced the record bits, which allow us to achieve the same function as the real-time controller.

The non-FT stage with the introduced record qubits is shown in the top half of Figure 2. This is the part of the circuit where stabilizer measurements p_3 , p_4 , and p_5 are made. All the record bits are initialized at $|0\rangle$, and each of the classical measurements of the auxiliary bit is coupled with a record bit through a CNOT gate. Therefore, when the classical measurement is done, the auxiliary bit collapses to the state recorded by the classical measurement i.e., if we measure $|0\rangle$ on the auxiliary bit, the record qubit will remain at $|0\rangle$; but if we measure $|1\rangle$ on the auxiliary bit, the CNOT gate controlled on the auxiliary bit after measurement will change record bit coupled to it and measure $|1\rangle$. Therefore, after the circuit finishes executing one cycle of the non-FT stage, the result of the classical measurements are effectively stored in the record bits such that $|r_0r_1r_2\rangle = |m_1m_2m_3\rangle$.

Now that the results of the stabilizer measurements are stored in the record bits, we find a way to access the bits after each cycle of the non-FT stage. The simulator in `Qiskit` gives the resulting state of all quantum bits in the following order: $|d_0d_1d_2d_3d_4a_0f_0r_0r_1r_2\rangle$ where d_i are the data bits, a_0 is the auxiliary bit, f_0 is the flag bit, and r_i are the record bits.

Thus, we can access the state of the record bits $|r_0r_1r_2\rangle$ by first performing a partial trace over the space of the data bits, auxiliary bit, and the flag bit (refer to Equation 2):

$$\rho_R = \text{Tr}_{D,A,F} [|d_0d_1d_2d_3d_4a_0f_0r_0r_1r_2\rangle \langle d_0d_1d_2d_3d_4a_0f_0r_0r_1r_2|]. \quad (2)$$

The result of Equation 2 gives the reduced density matrix of the record bits. Since the the state of the record bits $|r_0r_1r_2\rangle$ is guaranteed to be a pure state, we can get $|r_0r_1r_2\rangle$ from the reduced density matrix by simply reading off the matrix elements of ρ_R .

Now we can access $|r_0r_1r_2\rangle$ after each cycle. Per cycle, we check if $|r_0r_1r_2\rangle = |000\rangle$. If it is, the simulation terminates and the resulting state of the last cycle will be the prepared state; otherwise, another cycle of the non-FT stage is simulated.

After the simulation successfully terminates, meaning that all the stabilizer measurements returned $|0\rangle$, we get the states of the data bits again by performing a partial trace over the space of all other trivial bits.

A snippet of a pseudocode that executes the above steps (and acts as a real-time controller) is:

Success Rate For Preparing $|-\rangle_L$

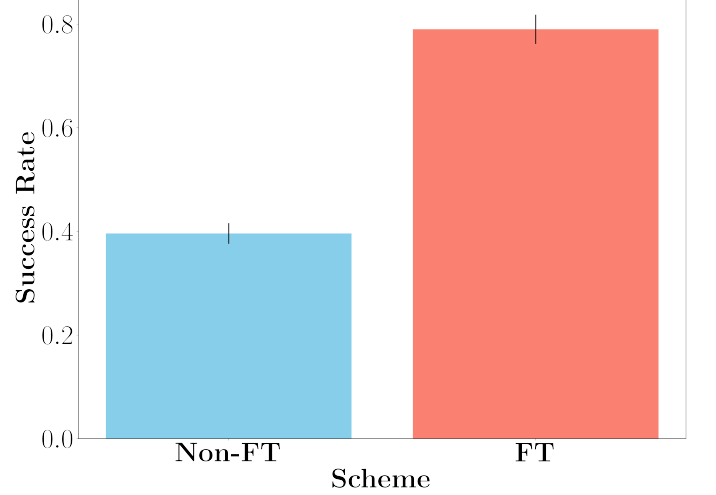


FIG. 4. The success rate of preparing $|-\rangle_L$ for non-FT and FT preparation

```

2 while record_state != '000':
3
4     # run one cycle of the sim on the non-FT stage
5
6     simulator = get_qiskit_simulator()
7     result_state = simulator.run(nFT_circuit)
8
9     # trace over the space of non-record bits
10
11     record_rho = partial_trace(result_state, [(0,6)])
12
13     # get result state by reading rho
14
15     record_state = get_state(rho)
16
17 # once the iteration stops,
18 # the result state is the desired state
19 # trace over the space of non-data bits
20
21 logical_state = partial_trace(result_state, [(5,9)])

```

Having achieved an equivalent of real-time control, we test the integrity of the prepared $|-\rangle_L$ by looking at its stabilizer measurements. We found that when there's no error, our version of the non-FT prepares the desired state with 100% probability, and the Pauli operator measurements of the prepared state are shown in Figure 3.

C. Fault-Tolerant Stage and Introduced Error

Having verified the successful preparation of $|-\rangle_L$ from the non-FT stage, we constructed the circuit with the FT stage. This is the bottom half of the circuit in Figure 2 where stabilizer measurements T_1 and T_2 are made. We constructed the circuit with the FT stage and coupled each classical measurement with a record bit using the

CNOT gate to achieve real-time control.

To compare the success rate of preparing $|-\rangle_L$ using the non-FT and FT circuits, we introduce Pauli errors onto the auxiliary bit in both the non-FT and FT stages. These Pauli errors are I, X, Z, Y with probabilities 0.4, 0.2, 0.2, 0.2 respectively in the non-FT and FT stage to compare

Note that the entirety of the circuit in Figure 2 was used to get the success rate of preparing $|-\rangle_L$ with the FT-stage, i.e., p_3, p_4, p_5, T_1 and T_2 were all used to determine the success rate. On the other hand, only p_3, p_4 , and p_5 were used to get the success rate of the non-FT stage. In other words, measurements m_{T_1} and m_{T_2} are only used to assess the fidelity of $|-\rangle_L$ in the FT stage.

D. Fidelity and Success Rates

We check the integrity of the prepared state by looking at its stabilizer measurements since the desired state is the unique +1 eigenstate of $p_1 = IZXXI$, $p_2 = ZIIZX$, $p_3 = XZIIZ$, $p_4 = ZXZII$, $p_5 = IIZXX$.

For both non-FT preparation and FT preparation, we ran 1000 iterations each and compared the resulting state with the desired $|-\rangle_L$ state. The success rate of preparing $|-\rangle_L$ is shown in Figure 4 where the error bars are the projected noise in sampling.

We see that the FT preparation has a much higher success rate compared to the non-FT preparation. The FT preparation provides a success rate of 79.0% compared to the non-FT stage's 39.6%. We have shown the benefit of an FT code in improving the success rate of preparing a desired logical state.

VI. CONCLUSION

- We explore a five-qubit quantum error-correcting code. We use an auxiliary qubit to perform re-

peated stabilizer measurements and a flag qubit to detect possible errors on the auxiliary qubit. We use record qubits to allow for non-destructive stabilizer measurements.

- Our circuit is divided into a non-fault-tolerant and a fault-tolerant stage.
- We demonstrate the fault-tolerant preparation of the logical state $|-\rangle_L = \frac{1}{\sqrt{2}}(|0\rangle_L - |1\rangle_L)$. $|-\rangle_L$ is the +1 eigenstate of $p_1 = IZXXI$, $p_2 = ZIIZX$, $p_3 = XZIIZ$, $p_4 = ZXZII$, $p_5 = IIZXX$. In the non-FT stage, we make stabilizer measurements p_3, p_4, p_5 . In the FT stage, we make stabilizer measurements $T_1 = p_2 p_4 p_5$ and $T_2 = p_1 p_3 p_5$. If $m_{T_1} = m_2 m_4 m_5$, $m_{T_2} = m_1 m_3 m_5$ and the flag is $|0\rangle$, we prepare $|-\rangle_L$ successfully.
- We introduce Pauli errors I, X, Z, Y with probabilities 0.4, 0.2, 0.2, 0.2 respectively in the non-FT and FT stage.
- We run simulations for 1000 iterations and compare the simulated stabilizer measurements (with possible errors) against the expected stabilizer measurements for $|-\rangle_L$. We find that the success rate of preparing $|-\rangle_L$ is 0.396 in the non-FT stage and 0.790 in the FT stage.

VII. APPENDIX

Our code to run Qiskit simulations described in this paper can be found here: C191 Fault Tolerant Prep.

[1] Abobeih, M. H., Wang, Y., Randall, J., et al. 2022, Nature, 606, 884–889, doi: 10.1038/s41586-022-04819-6
[2] Hanson, R., Gywat, O., & Awschalom, D. D. 2006, Phys. Rev. B, 74, 161203, doi: 10.1103/PhysRevB.74.161203

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[4] Javadi-Abhari, A., Treinish, M., Krsulich, K., et al. 2024, Quantum computing with Qiskit, doi: 10.48550/arXiv.2405.08810